



The Gainward GeForce FX 5900 is the budget gamer's delight

In the nVidia 5900 series we tested three brands, Gainward, Pixelview and MSI—all three performed superlatively. Though there wasn't much variation in the other benchmarks, the Gainward card lost out slightly on the UT2003 test. The competition was neck-and-neck between Pixelview and MSI cards, with both scoring similarly, but the former lost out due to a few missing features and lower performance score in Quake III—thus the MSI took the spoils in the 5900 group.

Of the three 5800 series cards we got,—the Asus V9900/TD, the Gainward Ultra/1000 Plus and the PixelView GeForceFX 5800—the Gainward Ultra/1000 Plus outperformed both the 5900 series, as well as the 9800 series of cards—getting higher scores in both performance, as well as value for money. Unfortunately, it heats up a lot and makes too much noise, which is why nVidia decided to discontinue it in the first place.

The ATi 9700 PRO chipset was the first card to support DirectX 9. ATi beat nVidia at its own game, and launched this, much to the surprise of the whole industry, in mid 2002. For a card that old, it packs a punch with an impressive score of 3857 marks in 3DMark 2003 at a resolution of 1024 x 768. It also scored a comfortable 28.4 fps in the GunMetal Benchmark—one of the most rigorous benchmarks we used. Unfortunately, the fact that ATi is phasing out the 9700 Pro chipset and replacing it with the 9800 series leaves this card out in the cold.

The real performer in this category was the Gainward GeForce FX 5900 Ultra (256 MB), it was the only card to cross the 130 fps mark in the SeriousSam benchmark in high-quality

64-bit Gaming

The most obviously noticeable aspect in today's high-end gaming world, is the complete disparity between the graphics subsystem and the general PC architecture. While the GPU runs at close to 128-bit architectures, the CPU and its supporting system has a 32-bit architecture. Thus there is always a bottleneck, no matter how fast your GPU is. All this seems to be on the verge of changing, with AMD launching its 64-bit desktop processor the Athlon 64 FX51 which promises to change the face of personal computing. Moreover, Apple has also launched its PowerMac G5 that is based on 64-bit computing. What does all this mean to the gaming world? These 64-bit architectures have many more registers on the CPU that make it possible to manipulate more data simultaneously. Each piece of data is double in size, at 64-bits instead of 32-bits, so this too provides boost. This drastically improves the performance of the CPU, thereby leaving the GPU to render much more heavy duty stuff. Moreover, the size of the bus on the motherboard is double that on the current 32-bit versions and thus, more bytes of data can be transported across various parts of the computer in the same clock cycle.

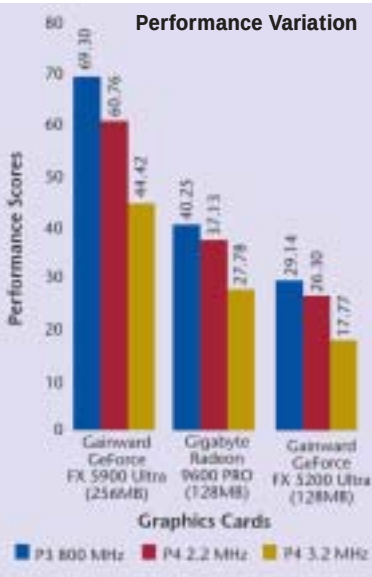
Epic Games has announced that they will be coming up with a 64-bit version of Unreal Tournament 2003 sometime next year, which will be an interesting test of the actual performance boost.



Decision Maker

Consider a low-end video card such as the Gainward GeForce FX 5200 Ultra for a PC running a PIII 800 MHz or a P4 1.3 GHz. Priced at around Rs 11,000; it pitched in a performance of 17.77 per cent on the PIII 800 MHz system.

Considering that the Gigabyte Radeon 9600 PRO scores 37 per cent and the Gainward GeForce FX 5900 Ultra pitches in 60 per cent on the P4 2.2 GHz and 69 per cent on the P4 3.2 GHz, opt for the latter in case price is not an issue. In the 1.4 to the 2.2 GHz range, we recommend the Gigabyte Radeon 9600 PRO since it gives a better price-to-performance ratio.



mode. It scored slightly higher than the MSI 5900 card thereby capturing the best performer award.

And the winner is...

Overall, the nVidia 5900-based cards and ATi 9800-based cards fought a valiant fight, but in the end the 5900 cards won. The Gainward GeForce FX 5900 Ultra took the best performance award and the Gainward 5900 took the best value award.

ATI has been trying to make a comeback in the graphics cards market. They have some excellent offerings in the mid-range market, but when it comes to the hardcore gamer who needs the best of graphics cards, ATI clearly loses out to nVidia. This is quite evident from the massive score difference between the 5900 and the 9800 PRO. However there is a caveat. The first real DirectX 9 game shows a possibly different picture.

A huge disappointment was the average performance by the ASUS cards. At one time, ASUS was one of the most reputed manufacturer of high-end graphics cards, but this doesn't hold true anymore. Overall, the graphics card scene looks very good with more vendors coming up with better products.

PRAVEEN KURUP
praveen_kurup@thinkdigit.com
BHASKAR BANIK
bhaskar_banik@thinkdigit.com

Answers to quiz from page no. 132:

Crossword:

Down:

1. Bhatkar
2. Ethernet
3. Bezos
4. Bhatia, 7. Gates
5. Analytical
6. Palmisano

Across:

1. BernersLee
2. Allen, 9. Mitnick
3. Capek
4. Bhatia, 7. Gates
5. Analytical
6. Palmisano

Answer:

1. c, 2. b, 3. a, 4. c, 5. a, 6. a, 7. a, 8. b, 9. a, 10. a, 11. b, 12. a, 13. b, 14. a, 15. b